

Piotr Koper

Data Engineering · Pipeline Architecture · Cloud Infrastructure

Senior Data Engineer specialising in platform engineering, IaC, and CI/CD, with 4+ years of experience building scalable data pipelines and infrastructure. Currently leading a working group across Terraform, GitLab CI/CD, and Docker image management.

ABOUT

I came into data engineering the long way round. Over a decade in logistics and operations management — Halfords, DHL, Aston Martin Lagonda — taught me how supply chains actually work, how to lead cross-functional teams under pressure, and crucially, how to build tooling when the tools you're given aren't good enough.

In 2021 I was selected for JLR's internal Data Engineering conversion programme, run with Infinity Works. It was competitive, it was intense, and it was the best decision I've made professionally. What followed wasn't a clean pivot into writing SQL — it was a gradual move toward the parts of the stack I find most interesting: infrastructure, automation, and the systems that make data teams actually function.

Four years in, I lead working groups across GitLab CI/CD pipelines, Docker image management, and Terraform at JLR. I own the Workload Identity Federation rollout between GitLab and GCP. I spend more time thinking about authentication patterns and Terraform module design than I do writing pipeline code — and I'm completely fine with that.

I'm interested in roles that sit at the intersection of data and platform engineering. Somewhere that values building things properly, documenting them well, and automating the things

BASED IN

West Midlands, UK

CURRENTLY

Senior Data Engineer, JLR

BACKGROUND

Logistics & Operations → Data Engineering → Platform Engineering

EDUCATION

MSc Information Systems,
Technical University of Łódź
MSc ICT for Engineers,
Coventry University

LANGUAGES

Polish — native
English — fluent

that shouldn't need a human. Rust is on the learning list.
Opinions on tab width are strong.

SKILLS & STACK

Infrastructure & IaC

Terraform GCP Docker Kubernetes Helm Artifact Registry Cloud Run
Workload Identity Federation

CI/CD & DevOps

GitLab CI/CD GitHub Actions GitLab Components Pre-commit
Container Scanning Pipeline Templating

Data Engineering

Dagster Apache Airflow dbt ETL / ELT Data Modelling
Pipeline Architecture

Languages

Python SQL Bash HCL YAML

Databases & Storage

PostgreSQL GCS BigQuery Cloud SQL

Observability & Tooling

Grafana Git Obsidian Linux uv pnpm

EXPERIENCE

2022 — Present Senior Data Engineer
Jaguar Land Rover

Part of the core Data Engineering platform team, focused on infrastructure, developer tooling, and engineering standards across the data organisation. Progressed from pipeline development into platform and infrastructure engineering, taking on technical leadership across several working groups.

CI/CD Working Group — Lead

Lead the GitLab custom CI/CD pipelines working group, responsible for refactoring, improving and standardising pipeline templates across the data engineering organisation. Designed a reusable authentication template supporting both Workload Identity Federation and legacy Service Account Key methods, ensuring backward compatibility across all existing projects.

Workload Identity Federation — Owner & Lead

Sole owner of the WIF rollout between GitLab and GCP. Designed and provisioned the full infrastructure via Terraform — Pool, Provider and bindings — and redesigned the GitLab CI/CD authentication layer to support WIF while remaining fully compatible with existing projects. Fully documented end to end.

Docker Images Working Group — Lead

Lead the working group responsible for designing, maintaining and building custom Docker images stored in GCP Artifact Registry, used across data engineering pipelines and services.

Terraform Working Group — Lead

Recently took on leadership of the Terraform working group, overseeing module standards, provisioning patterns and infrastructure-as-code practices across the data platform.

Notifications System

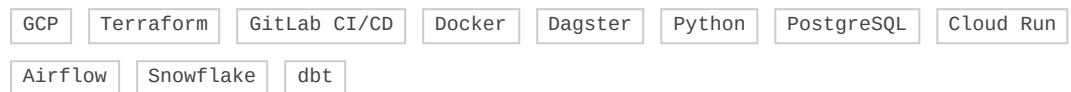
One of the core developers of a GCP-based notifications system that triggers messages to the right channels based on subscription behaviour and API endpoint calls during pipeline execution. Fully refactored the Terraform module for Cloud Run and PostgreSQL infrastructure, and authored extensive documentation covering database management and operational runbooks.

GitLab Instance Migration

Part of the team that designed and provisioned the Kubernetes cluster, namespace and Helm module for GitLab runners during a full GitLab instance migration — infrastructure managed entirely via Terraform.

Python Task Orchestrator POC — Lead

Led the evaluation of enterprise Python task orchestrators, assessing Astronomer and Dagster against defined criteria. Produced a technical evaluation report adopted as the recommendation to the Enterprise Technology Board — Dagster selected as the preferred platform.



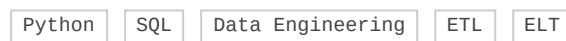
2021 — 2022

Data Engineering Conversion Course

[Academy](#) · [Infinity Works](#) · [Jaguar Land Rover](#)

Successfully selected for a competitive internal conversion programme, transitioning into data engineering from a non-engineering background. Completed the structured curriculum covering data pipelines, ETL and ELT patterns, SQL, and Python — formally completed May 2022.

This programme marked the start of a deliberate career transition into data engineering, building the foundations that underpinned subsequent platform and infrastructure work.



2018 — 2021

CPAT Lead & Planner

[Jaguar Land Rover](#)

Cross-functional programme role managing the Capacity, Parts Availability and Timing assessment process across JLR vehicle programmes. Responsible for critical path planning, risk management, and stakeholder communication across Purchasing, Supply Chain and Programme teams.

Built several internal tools to address existing process gaps — including a redesigned CPAT Input Matrix to improve data control and an automated WIP file creation process saving approximately 12 hours per file. Also developed an eCAR data scrape tool integrated directly into the WIP workflow.

This role developed the stakeholder management, cross-functional leadership and data-driven decision making skills that directly underpin the working group leadership taken on in data engineering.



2007 — 2018

Planning Manager & Logistics Operations

Halfords · Aston Martin Lagonda (DHL) · DHL

Over a decade of progressive logistics and operations management across major UK distribution and manufacturing environments, including Halfords (1.4M units/week), Aston Martin Lagonda, and DHL.

Consistently built tooling and reports to replace manual processes — a Logistics Performance Tool tracking volumes, hours and costs against budget; automated planning tools; and redesigned operational workflows. The throughput numbers and team sizes involved meant data quality and analytical rigour weren't optional — they were how the operation ran.

Selected for a competitive DHL Head Start management programme from hundreds of UK and Ireland applicants, one of 28 accepted.

This background directly informs how I approach data engineering — the domain problems, the stakeholder relationships, and the instinct to automate anything done more than twice.

Excel

VBA

Operations Management

Data Analysis

PROJECTS

Octopus Energy Dashboard

Built a personal data pipeline to track and visualise energy costs and consumption using the Octopus Energy API. The project covers the full data lifecycle — ingestion, transformation, storage, and visualisation.

Pipeline

Dagster-based pipeline runs multiple times daily, pulling variable tariff rates and consumption data from the Octopus API, performing cleaning and transformation in Python, and writing to a PostgreSQL database hosted on an Oracle Cloud VM.

Infrastructure & Deployment

GitHub Actions handles deployment, authenticating to the Oracle VM and deploying the Dagster pipeline automatically. The database is connected to a Grafana dashboard providing day-on-day and week-on-week comparisons, including year-over-year cost tracking.

Why it matters

Built entirely outside of work, this project touches every layer of a real data platform — orchestration, transformation, storage, deployment and visualisation — using the same tools evaluated professionally (Dagster) and hosted on real infrastructure.

Dagster

Python

PostgreSQL

Grafana

GitHub Actions

Docker

EDUCATION & CERTIFICATIONS

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| 2007 | DEGREE
MSc Management and Production — Systems and Information Technologies
Technical University of Lodz, Poland |
| 2006 | DEGREE
MSc Information and Communication Technology for Engineers
Coventry University, United Kingdom |
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CONTACT

Open to data and software engineering roles with a focus on infrastructure, platform engineering, and developer tooling. Equally comfortable in CI/CD, IaC, containerisation, and data systems — interested in roles that value engineering depth over narrow specialisation.

piotr.koper.v0@gmail.com